Rotaract Club of Xiamen University Malaysia

Rotary Club of Kuala Lumpur Dynamics

Rotary Club of Tun Razak Exchange

SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named “SDC2026_TeamName_Preliminary.pdf” before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

Instagram: @sdc_xmumhack ***Email:*** sdc.xmumhack@gmail.com

1. TEAM AND PROJECT OVERVIEW

Team Name

DuoLogic

Selected Domain

Domain 1

Selected SDG Track (1–17)

4,5,8,9,10

Project / Solution Name

Nexora

Name (Leader)

Chang Zhi Qing

Email & Phone (Leader)

changzhiqing1996@gmail.com (0192723941)

University Affiliation(s)

University Tunku Abdul Raman Sungai Long

Number of Team Members

2

2. EXECUTIVE SUMMARY & PROBLEM STATEMENT

2.1. The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

AI adoption is changing the skills employers need faster than workers can updatetheir career plans. The Domain 1 brief reports that 56% of Malaysian professionalsuse AI at work, but only 26% have received formal training (Hays Asia Salary Guide 2025, as cited by Aiversecity). It also cites a Malaysian government estimate that over 600,000 workers will need reskilling within three to five years

Our primary users are Malaysian final-year students, fresh graduates and early-career workers who are unsure which skills will remain valuable. Existing career tools often match a CV to current vacancies; they rarely explain how a changing AI landscape affects future roles or provide an affordable, actionable learning sequence.

This creates risks of unemployment, skills mismatches and unequal access to opportunities. Nexora helps users identify transferable strengths, explore emerging roles and choose evidence-linked reskilling options. It aligns especially with SDG 4(quality education) , SDG 8 (decent work), SDG 9 (innovation), SDG 5 (genderequality) and SDG 10 (reduced inequalities

2. EXECUTIVE SUMMARY & PROBLEM STATEMENT

2.2. The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

Nexora is a browser-based workforce-planning platform for individuals and organisations. A person registers through Clerk and builds a profile with skills, experience and interests, or uses Discover to analyse a DOCX/TXT/MD CV or answer a questionnaire. CV text is parsed locally; PDF CVs are not supported by the current reader. Rule-based algorithms score career directions using skills (70%) and interests (30%), and show matched/missing skills. A separate scenario view changes the weights:
moderate change uses 65% skills, 20% interests and 15% live job signals; faster disruption uses 50%, 15% and 35%. These are transparent planning assumptions, not AI-generated forecasts.

An Express API retrieves dated remote job advertisements from the public Himalayas API, caches them and counts skill/AI mentions. The app shows source attribution, refresh times and limitations.
Users can view matching vacancies, links to employers, personalised learning steps, estimated hours, external course providers, self-reported progress and SDG tags. A risk view references US Bureau of Labor Statistics occupational projections. Additional organisation features include administrator-led teams, skill-gap matrices and optional synthetic test data.
The architecture uses React/TypeScript/Vite, Node/Express, PostgreSQL with Drizzle ORM, and Clerk authentication. A Help chat can optionally call Google Gemini or Anthropic Claude when an API key is configured; otherwise it uses built-in FAQs. The matching engine does not dependon a language model. Deployment to Replit is documented, but a live production URL was not supplied in the archive.

3. TECHNICAL ARCHITECTURE & IMPLEMENTATION PLAN

3.1. Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Frontend: React 19, TypeScript 5.9, Vite 7, Tailwind CSS 4 and accessible UI components.
Backend: Node.js 24, Express 5; REST endpoints for profiles, market signals, vacancies, team
planning and Help.
Data: PostgreSQL and Drizzle ORM; account-owned profiles, progress, organisation/team data and
synthetic demo employees.
Auth: Clerk sign-in, verified sessions and server-side ownership checks.
External sources: public Himalayas remote-jobs API (cached 6 hours, timestamps shown); cited US
BLS projections; external training-provider links.
Optional: Gemini or Claude only for Help chat; SMTP for invitation emails.
Deployment: Replit configurations included; a separate verified public deployment has not been shown.
Flow: browser -> Express REST API -> PostgreSQL/Clerk; API -> Himalayas -> dated skill signals
-> browser matching and plans.
Official SDC sponsor-provided services: Replit, provided with the promotion code.

3. TECHNICAL ARCHITECTURE & IMPLEMENTATION PLAN

3.2. Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

Empathise and define: prioritise a worker or graduate deciding which skills to learn next, rather than an open-ended career chatbot. Design a short profile -> market signals -> explained roles -> learning steps journey.
Prototype and test: the source implements CV-text extraction, questionnaires, role scoring, a live Himalayas-powered market view, SDG mapping, moderate/ fast scenarios, learning plans and progress tracking.

Test profile creation, skill extraction, role ordering,visible skills gaps, links, stale -source failures, mobile navigation and authorised access.
Measure: time for a new test user to finish the journey, proportion of recommendations with links and dates, and whether changed trend data changes scores. Iterate after feedback.

4. PRELIMINARY ENVIRONMENTAL & SDG IMPACT FRAMEWORK

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess Innovation and Creativity as well as Feasibility of build. Vague or unsupported projections may weaken your case for advancing to Phase 2.

4. PRELIMINARY ENVIRONMENTAL & SDG IMPACT FRAMEWORK

Metric Category	Your Response
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	SDG 4.4: increase the number of youth and adults with relevant employment and entrepreneurship skills by 2030. Supporting: SDG 4.3: equal access to technical and vocational education; SDG 8.5: productive employment and decent work.
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	56% of professionals use AI at work but only 26% received formal training (Hays Asia Salary Guide 2025, cited in domain brief). These are different measures: a 30-percentage-point adoption/training gap, not a directly observed untrained-worker rate.

4. PRELIMINARY ENVIRONMENTAL & SDG IMPACT FRAMEWORK

Metric Category	Your Response
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided*	% of profile walkthroughs that produce,within 3 minutes, ranked roles, skill-gap explanations, at least one cited current market source, and an ordered learning pathway with SDG tags. Also track completion time in minutes.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	>=80% of at least10 user walkthroughs complete this full process in <=3 minutes, and 100% of shown live-signal panels identify their source and date.

5. ELIGIBILITY & READINESS CHECKLIST

Tick all boxes that apply before submitting.

- ☑ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- ☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- ☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- ☑ All fields in this template have been completed within the word limits stated.

6. IP & ORIGINALITY DECLARATION

6.1. Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☑ Yes ☐ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

the Nexora source archive (React frontend, Express API, Drizzle/PostgreSQL schema, Clerk integration and user-interface components) existed by
22 Oct 2026.

6. IP & ORIGINALITY DECLARATION

6.2. Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☑ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
React / Vite / TypeScript	Code (web UI / build)	MIT (React, Vite); Apache-2.0 (TypeScript)
Express / Drizzle / PostgreSQL	Code / DB packages	MIT (Express); Apache-2.0 (Drizzle); PostgreSQL License
Clerk SDK	Code / authentication	MIT (SDK)
Himalayas jobs; US BLS	Public job data / statistics	Himalayas API Terms of Use / BLS Public Domain (with exceptions)
Radix UI / Lucide	Code and icons	MIT / ISC

6. IP & ORIGINALITY DECLARATION

6.3. Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☑ No

If yes, please describe the nature of the agreement and which member it concerns:

7. DECLARATION

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

[Signature]

Zhi

[Date]

23/10/2026